

1. Which one of the following is the ultimate goal of financial management?
 - A. Reduce costs
 - B. Reduce SG&A expenses
 - C. Increase net income as much as possible
 - D. Create value for stockholders by maximizing the stock price

2. The correct statement is:
 - A. The accounting statement that measures the revenues, expenses, and net income of a firm over a period of time is called cash flow statement.
 - B. The accounting statement that summarizes a firm's accounting value, changes in cash flow, and gross PPE as of a particular date is called a balance sheet.
 - C. The accounting statement that measures the revenues, expenses, and net income as of a particular date is called a balance sheet.
 - D. The accounting statement that measures the revenues, expenses, and net income for a certain period of time is called an income statement.

3. The **incorrect** statement about the agency issue is:
 - A. It refers to the conflict of interest between a principal (CEO) and an agent (stockholders).
 - B. A principal hires an agent to represent his/her interests but the agent may act on his/her own interest at the expense of a principal's benefit.
 - C. The agency issue not only exists in publicly listed firms but also exist in privately held firms as long as the owner of a firm hires a CEO to manage the firm.
 - D. Managers like to buying other firms to build an "empire", even some firms they bought have bad financial performance. This is because when managers manage a bigger company and more people, they then can ask shareholders for more compensation. This is an example of the agency issue.

4. Which of the following is **NOT** a good way to mitigate the agency issue? A. Managerial compensation and bonus can be used to align management and stockholder interests.
B. Give managers stocks so that managers become the shareholders as well C. Threat of a takeover or merger and acquisition can be one of the ways to reduce the agency issue.
D. Do not hold a conference call after a firm reports its quarterly earnings so that managers can hide some important information.
5. **(challenging question)** The **incorrect** statement about the Sarbanes-Oxley Act (or SOX) is_.
A. It was passed by the US Congress in 2002.
B. The SOX Intended to strengthen protection against accounting fraud and financial malpractice
C. Compliance are very costly and some firms were driven to go public outside the US or simply go dark (go private).
D. The SOX came in response to financial scandals in the internet bubble or the dotcom bubble, in the late 1990s.

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6. Tom earned \$120 in interest on his savings account of \$120,000 last year. Tom has decided to leave the \$120 in his account so that he can earn interest on the \$120 this year. That is, he will NOT withdraw any money from his account. This process of earning interest on prior interest is called:
A. simple interest
B. compounding or compound interest
C. present value
D. discountfactor
7. Which one of the following is the correct way to compute the present value of \$600 to be received in 6 years (or at the end of the 6th year)? The discount rate is 7 percent. A. $PV = \$600 (1 + 6)^7$
B. $PV = \$600 (1 + 0.07)^6$
C. $PV = \$600 \times (0.07 \times 6)$
D. $PV = \$600 / (1 + 0.07)^6$
8. Today at 1/1 2021, you deposit \$2,400 in a bank account that pays 4 percent **simple interest**. How much total interests will you earn over the next 5 years (at 1/1/2026)?
A. \$96.00
B. \$101.15
C. \$480.00
D. \$492.16
9. Angela has just received an insurance settlement of \$35,000 at the beginning of the year. She wants to save this money until her daughter goes to college. If she can earn an average of 5.5 percent, compounded annually, how much will she have saved when her daughter enters college 10 years from now?
A. \$48,000.00

- B. \$50,929.02
- C. \$59,875.06
- D. \$59,785.06

10. **(challenging question)** You just won \$50,000 and deposited your winnings into an account that pays 5.5 percent interest, compounded annually. How long will you have to wait until your winnings are worth \$100,000?
- A. 7.24 years
 - B. 8.00 years
 - C. 9.29 years
 - D. 12.95 years
11. You want to have \$25,000 for a down payment on a house 6 years from now. If you can earn 6.5 percent, compounded annually, on your savings, how much do you need to deposit today to reach your goal?
- A. \$17,133.35
 - B. \$17,420.73
 - C. \$17,880.69
 - D. \$18,211.17

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12. You are offered an opportunity to buy an investment product by a banker that will pay you •
- \$200 in year 1,
 - \$400 the next year,
 - \$600 the following year, and
 - \$800 at the end of the 4th year.

Assume you can earn 10% on similar investments (that is, your opportunity cost is 10%). The banker asks you to pay \$1600 to buy this product. Are you going to accept the offer or not? A. NO, because the PV of this series of cash flows is \$1,432.93, which is the most that I should pay to purchase this product.

- B. YES, because I totally receive \$2000 in four years.
- C. NO, because if purchase this product my return is 15%, which is higher than my opportunity cost.
- D. NO, because the PV of this series of cash flows is less than \$1600.

13. Which of the following publicly listed firm has the largest market cap in the world? A.
- TSLA
 - B. NVDA
 - C. MSFT
 - D. AAPL

14. NKLA is one of the stocks that you do not like and expect its price to drop. So you short 1000 shares of NKLA at the price of \$25 per share. The correct statement is . A. This is a go short transaction; that is, you first borrowed \$25,000 from your broker and later you need to return \$25,000 to your broker.
- B. Assume the maximum loss you can sustain is 20%. When the stock price drops to \$20, you

incur this maximum loss of 20%.

C. The maximum profit of this transaction could be \$25,000 but the maximum loss could be infinite.

D. You can use a limit buy order to protect yourself from the maximum loss.

15. **(challenging question; pls google, do some research, or watch my video on bitcoin uploaded at CANVAS FILE)** According to WSJ, Tesla Inc. said Monday 2/8 that it bought \$1.5 billion in bitcoin, a disclosure that follows Chief Executive Elon Musk's promotion of the cryptocurrency and other digital-currency alternatives on Twitter. The electric-vehicle company also said it expects to start accepting bitcoin as payment for its products soon. Bitcoin prices jumped more than 10% after the announcement, according to cryptocurrency research and news site CoinDesk. The purchase of \$1.5 billion in bitcoin is booked as on Tesla's .

A. cash and cash equivalents; income statement

B. intangible asset; balance sheet

C. investments; balance sheet

D. other investments; income statement

16. The IRR (internal rate of return) is the discount rate that makes the NPV of all cash flows (including the initial capital outlay) .

A. >0

B. $=0$

C. ≥ 0

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D. $>$ interest rate

17. STOCKA reports 2020 EPS (earnings per share) of \$9.00 in its annual report (10K). Stock analysts on average estimated its EPS to be \$10 (this is the market expectations). STOCKA stock price as of 9/27/2021 is \$135. Its TTM PE multiples are; generally speaking, STOCKA can be considered a stock.

A. 15%; value

B. 15; growth

C. 15; value

D. 15%; expensive

18. The incorrect statement about NPV method is:

A. It considers all cash flows

B. It considers time value of money

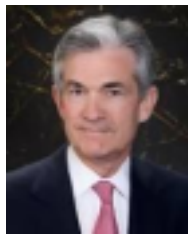
C. It adjusts for risk

D. It cannot rank mutually exclusive projects

19. Who is the current Chairperson of Fed Reserve Bank of the US? When s/he speaks, the whole world listens.



A.



B.



C.



D.

Questions 20 to 28 are based on the following financial statements of Marc & Martin Company. Note it has a tax rate of 15%.

INC		
	2019A	2018A
Sales	600.0	460.0
COGS	250.0	220.0
Gross Profit	350.0	240.0
SG&A	66.0	58.0
R&D	14.0	12.0
Depreciation exp	40.0	35.0
EBIT	230.0	135.0
Interest Exp.	30.0	25.0
Pre-tax earnings	200.0	
Taxes	30.0	16.5
NI		

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BS		
	2019A	2018A
Cash & cash equivalents	100.0	100.0
AR	300.0	250.0
Inventory	400.0	300.0
Current Asset	800.0	650.0

Gross PPE	4000.0	3600.0
Accu. Dep.	1900.0	1860.0
Net PPE		1740.0
Total assets		2390.0
AP	200.0	200.0
Current portion of LT debt	200.0	100.0
Current liabilities	400.0	
LT debt	1140.0	900.0
Total liabilities		1200.0
Common Stock	500.0	500.0
Retained Earnings	860.0	690.0
Total equity		1190.0
Total liabilities & total equity	2900.0	2390.0

20. What is the company's NI (net income) in 2018A?

- A. \$49
- B. \$93.5
- C. \$170
- D. \$135

21. What is the company's net PPE (property, plant and equipment) at the end of 2019A?

- A. \$2100
- B. \$210
- C. \$4000
- D. None of the above

22. What is the company's TA (total assets) at the end of 2019A?

- A. \$1,360
- B. \$2,900
- C. \$2,390
- D. None of the above

23. What is the company's gross margin in **2019A**? Operating margin in **2019A**?

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- A. Gross margin = 38.33%; operating margin = 58.33%
- B. Gross margin = 58.33%; operating margin = 38.33%
- C. Gross margin = 28.33%; operating margin = 58.33%
- D. None of the above

24. Among the four margins, gross margin, operating margin, net profit margin, and R&D margin, which one is the MOST important one to help assess the company's profitability? Why? A. gross margin; because most stock analysts use this ratio.

- B. operating margin; because it measures the profitability generated from a company's core or major business.
- C. net profit margin; because it measures the net profit or net earnings after all the expenses being considered.
- D. R&D margin; because research and development are the key to a company's innovation.

25. What is the company's total long-term debt at the end of 2019A? What is the company's leverage (or debt ratio) at the end of 2019A?

- A. LT debt = \$1,140 ; debt ratio = 39.31%
- B. LT debt = \$1,340 ; debt ratio = 46.21%
- C. LT debt = \$400 ; debt ratio = 13.79%
- D. None of the above

26. What is the company's interest coverage ratio in 2019A?

- A. 7.67
- B. 20
- C. 5.67
- D. None of the above

27. (calculation problem). What is Marc & Martin Company's ROE in **2019**? Verify DuPont Equation. This is a calculation problem based on the financial statements above. You must show your calculations.

28. (calculation problem). Assume that Marc & Martin Company has working capital investment (WCInv) in 2019 of \$150. **That is, you do not need to estimate its WCInv.** Estimate its free cash flow in 2019. This is a calculation problem based on the financial statements above. You must show your calculations.

29. (calculation problem). Marc & Martin Company wants to determine whether or not to build a new factory in Austin, Texas in Jan. 2021. After extensive research work, the CFO has provided the following information:

- The company need to make a capital investment of 180,000 in Jan. 2021 (this is cash outflow). • At the end of the first year, they expect to receive cash inflow of 63,000; with a net income of 13,620;
- At the end of the second year, they expect to receive cash inflow of 78,000; with a net income of 63,300;
- At the end of the third year, they expect to receive cash inflow of 96,000; with a net income of 91,100.

Q: You are the CEO. The required rate of return on this investment project is 15%. Do you approve/accept this project or not based on IRR method? Why? That is, what is the return from this project based on the above information (using Excel)? **Note: you must draw the timeline for this problem. If you do not draw the timeline, you will be subtracted by 5 points.**

30. (essay question).

- a. List at least three new things that you have learned from this class so far. These new things **must have improved your critical thinking**. You need to discuss how they improve your critical thinking. You must type your answers using WORD. MINIMUM: 100 words.
- b. Generally speaking, do you think you can take a high level of risk? Or a low level of risk when making investment? Why? Given the risk level you can take, recommend a stock that you want to go long and justify. You must type your answers using WORD. MINIMUM: 100 words.

31. **(optional question)**. Do you believe this exam is easy, fair, or difficult? Do you believe the problems on this exam is consistent with our class discussions? Do you have any suggestions to the class? Feel free to share your comments. Thank you.

